University of Ottawa

Department of Mathematics and Statistics

MAT 2143B : Introduction to Group Theory

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First Midterm Exam

March 2, 2022

Surname _____ First Name _____

Student # _____ DGD _____

Instructions:

(a) You have 80 minutes to complete this exam and submit your PDF file on BrightSpace.

(b) All work to be considered for grading should be written on your solution sheets. You should clearly indicate where the solution of each question is written.

(c) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(d) You are strongly recommended to write in **pen**, not pencil.

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	3	4	5	3	4	4	23
Grade							

1. Which of the following numbers are co-prime to n for every $n \in \mathbb{N}$? You should justify your answer.

(a) **[1 point]** $n^2 + n + 2$

(b) **[1 point]** $n + 3$

(c) **[1 point]** $2n + 1$.

Solution:

(a) If n is even, for example $n = 2$, both n and $n^2 + n + 2$ are divisible by 2.

(b) If n is multiple of 3 then so is $n + 3$.

(c) Assume that $d|2n + 1$ and $d|n$. Then from $n|2n$ it follows that $d|2n$, and since $2n + 1 - 2n = 1$ we obtain $d|1$ hence $d = \pm 1$. Thus $2n + 1$ and n are always coprime.

2. (a) **[2 points]** Write down the definition of a group.

Solution: This is Def. 2.15 from the notes.
A *group* is a set G equipped with a binary operation $\star : G \times G \to G$ which satisfies the following:

- The binary operation $\star$ is *associative*, i.e., $(a \star b) \star c = a \star (b \star c)$ for all $a, b, c \in G$.
- G has an *identity element*. That is, there exists an element $e \in G$ such that $a \star e = e \star a = a$ for all $a \in G$. Sometimes e is called the *neutral element* of G.
- Every $a \in G$ has an *inverse*, i.e., for every $a \in G$ there exists an element $a^{-1} \in G$ such that $a \star a^{-1} = a^{-1} \star a = e$.

(b) **[2 points]** Write down all of the elements of the group $U(30)$. You should justify your answer.

Solution: The elements of $U(30)$ are the numbers from 1 to 30 that are co-prime to 30. Since $30 = 2 \times 3 \times 5$, we basically need to remove the multiples of 2, 3, and 5¿ Thus

$$U(30) = \{1, 7, 11, 13, 17, 19, 23, 29\}$$

3. (a) **[1 point]** Write down the definition of the order of an element of a group G.

Solution: This is Def. 6.3 from the notes. The order of $g \in G$ is the smallest $k \in \mathbb{N}$ such that $g^k = e$.

(b) **[2 points]** Which elements of $\mathbb{Z}_{48}$ have order 4? Determine all such elements. You should justify your answer.

Solution: Suppose $o(g) = 4$ for $g \in \mathbb{Z}_{48}$. Then $4g = 0$ in $\mathbb{Z}_{48}$, hence $48|4g$, so

$$4g = 48d \text{ for some } d \in \mathbb{Z}.$$

It follows that $g = 12d$, hence g can be one of $0, 12, 24, 48$. However, we should exclude 0 and 24 because $o(0) = 1$ and $o(24) = 2$. Thus the final answer is $\{12, 36\}$.

(c) **[2 points]** Write down the proof of the following statement: let G be a group and let $g \in G$ such that $o(g) < \infty$. Then for every $k \in \mathbb{Z}$ we have $g^k = e$ if and only if $o(g)|k$.
Solution: This is Theorem 6.6 from the notes.

Set $n := o(g)$. By the division algorithm we can write $k = nq + r$, where $0 \leq r \leq n - 1$. Note that

$$n|k \text{ if and only if } r = 0.$$

Now

$$g^k = g^{nq+r} = g^{nq}g^r = (g^n)^q g^r = e^q g^r = g^r.$$

If $r = 0$ then $g^k = e$, and if $1 \leq r < n$ then $g^k = g^r \neq e$ because $1 \leq r < n = o(g)$.

4. **[3 points]** Let $G = \mathbb{Z} \times \mathbb{Z}_4$ and let

$$H := \{(3a, 0) \,:\, a \in \mathbb{Z}\} \cup \{(3b, 2) \,:\, b \in \mathbb{Z}\}.$$

Is H a subgroup of G? You should justify your answer.

Solution: Yes, H is a subgroup. We verify this by the Concise Subgroup Test. choose $g, h \in H$. We need to verify that $gh^{-1} \in H$. There are 4 cases to consider:

I) $g = (3a, 0)$ and $h = (3a', 0)$. Then $h^{-1} = (-3a', 0)$, hence $gh^{-1} = (3(a - a'), 0)$, so $gh^{-1} \in H$ because $-a' \in \mathbb{Z}$.

II) $g = (3a, 0)$ and $h = (3b, 2)$. Then $h^{-1} = (-3b, 2)$, hence $gh^{-1} = (3(a - b), 0 - 2) = (3(a - b), 2)$, so $gh^{-1} \in H$ because $a - b \in \mathbb{Z}$.

III) $g = (3b, 2)$ and $h = (3a, 0)$. Then $h^{-1} = (-3a, 0)$, hence $gh^{-1} = (3(b - a), 2)$, so $gh^{-1} \in H$ because $b - a \in \mathbb{Z}$.

IV) $g = (3b, 2)$ and $h = (3b', 2)$. Then $h^{-1} = (-3b', 2)$, hence $gh^{-1} = (3(b - b'), 2 - 2) = (3(b - b'), 0)$, so $gh^{-1} \in H$ because $b - b' \in \mathbb{Z}$.

5. Let $\sigma \in S_{10}$ be the permutation given as follows:

$$\sigma = (4\ \ 7\ \ 5\ \ 2\ \ 8)(3\ \ 8\ \ 1\ \ 10)(9\ \ 4\ \ 5).$$

(a) **[3 points]** Express σ as a product of disjoint cycles.

(b) **[1 point]** Express σ^3 as a product of disjoint cycles.

Solution: (a) First we compute the array notation of σ:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 10 & 8 & 4 & 2 & 9 & 6 & 5 & 1 & 7 & 3 \end{pmatrix}$$

Thus the orbits of σ are

$$\{1, 10, 3, 4, 2, 8\} \text{ and } \{5, 9, 7\}$$

Therefore

$$\sigma = (1\ \ 10\ \ 3\ \ 4\ \ 2\ \ 8)(5\ \ 9\ \ 7).$$

(b) By (a) we have $\sigma = \tau_1\tau_2$ where τ_1 and τ_2 are disjoint cycles. Since disjoint cycles commute, we have $\sigma^3 = \tau_1^3\tau_2^3$. Now $(5\ \ 9\ \ 7)^3$ is the identity permutation and

$$(1\ \ 10\ \ 3\ \ 4\ \ 2\ \ 8)^3 = (1\ \ 4)(10\ \ 2)(3\ \ 8).$$

Therefore $\sigma^3 = (1\ \ 4)(10\ \ 2)(3\ \ 8)$.

6. [**4 points**] For each of the following statements, write **T** if the statement is true, and write **F** if the statement is false. You will receive 1 point for a correct answer, and -.5 points for an incorrect answer. (If you leave the answer space blank, you will not receive any negative points.) You do not need to justify your answers.

____ A non-empty subset of a finite group G that is closed under the binary operation of G is a subgroup of G.

____ The group S_3 is isomorphic to the group of symmetries of a regular hexagon.

____ If G is a group and H and K are subgroups of G, then $H \cup K$ is always a subgroup of G, but $H \cap K$ might not always be a subgroup.

____ If $k \in \mathbb{Z}_n$ is an element of order n then k is a generator of $\mathbb{Z}_n$.

Solution: T, F, F, T.